

Documentation (Roman Urdu)

Yeh file application ke RAG (Retrieval-Augmented Generation) engine ka "Tukrey karne wala (Chopper)" hissa hy.

File Ka Maqsad (Purpose)

AI ko jab 100 pages ki PDF parhani hoti ha tou ek sath puri pass nhi ho sakti (Token limit ar memory wjah sy). Yeh file puri PDF ko chotay chotay hisson (e.g. 800 characters) mein kaatti (chunk) karti ha tke Vector search ar embedding asaan ho jaye.

Code Blocks aur Functions ki Tafseel

1. `class ChunkingEngine`

Tafseel: Ye engine 3 kisam ki advanced strategies (tareeqay) isstml krta hy katne ky lehe:

- **Structural Chunking** `_structural_chunk()`: Agar Word file ya PDF main barey chotay "Headings" hn, tou ye un headings ky mutabiq kaattitahy tky meaning khrab n hou.
- **Paragraph Chunking** `_paragraph_chunk()`: Agar simple text hy tuo lines / spaces (`\n\n`) ko dekhty uway paragraph limits per kaatti ha.
- **Sliding Window Overlap**: Jb chunk kaatta jata h tu uske achri alfazou ka aagy wly chunk m (overlap=150) k lhaz sy rakha jta hz taakeh context loss na hyjye (Parih ka tasasssul nb tooty).

2. Parent-Child Relationship

- Yey har chunk klyh ek `parent_chunk_id` index preserve krtaa ah taak future mein AI wapis original location file ka talash karr sky!